

# Milestone 2

## Team task breakdown and contributions

Member 1 (Nitul Patel) data collection and preprocessing were done. This involved loading the datasets, missing values, duplication checking, and preparation of data to analyze them. The member further standardised the sentiment labels of the primary dataset and made sure that both the data sets were correctly formatted and prepared to further processing.

Member 2 (Nikhil Patel) Analysis and visualization of data were carried out. This involved carrying out an exploratory data analysis (EDA), drawing pie charts, histograms, scatter plots, and line charts among other forms of graphs, as well as, interpreting the report. The member also found patterns, trends, and associations in the data and worked in writing the insights and findings section.

## Justification and discussion of preprocessing techniques

The preprocessing of the two datasets was performed with careful consideration prior to the process of starting the analysis to enhance the quality of data and guarantee proper outcomes. This was done by first identifying the missing values and the missing records were eliminated before incorrect analysis could occur. The duplicate data also was scrutinized and eliminated to ensure that the dataset lacks the repetitive information. This was necessary to ensure consistency and reliability of the data.

The sentiment labels in the primary dataset were also standardized by changing the values like good and bad into Positive and Negative. This was so as to bring uniformity and make the data easy to analyze. The textual information was also purified of any useless symbols, links and additional spaces making it more readable and easier to analyze.

In the secondary data, numerical variables, including coding hours, the hours of AI use, and other variables, were to be verified to be in the right format. Any wrong or random values were taken care of. The preprocessing steps were required so that the datasets could be clean, structured and could be analyzed and this helped in coming up with meaningful and accurate insights.

```
import pandas as pd

df1 = pd.read_csv("pdata.csv")
df2 = pd.read_csv("sdata.csv")

df1.head()
```

| Unnamed: 0 | tweets                                                |
|------------|-------------------------------------------------------|
| labels     |                                                       |
| 0          | 0.0 ChatGPT: Optimizing Language Models for Dialog... |

```

neutral
1          1.0  Try talking with ChatGPT, our new AI system wh...
good
2          2.0  ChatGPT: Optimizing Language Models for Dialog...
neutral
3          3.0  THRILLED to share that ChatGPT, our new model ...
good
4          4.0  As of 2 minutes ago, @OpenAI released their ne...
bad

df2.head()

```

|   | hours_coding | coffee_intake_mg | distractions | sleep_hours | commits |
|---|--------------|------------------|--------------|-------------|---------|
| 0 | 5.99         | 600.0            | 1.0          | 5.8         | 2.0     |
| 1 | 4.72         | 568.0            | 2.0          | 6.9         | 5.0     |
| 2 | 6.30         | 560.0            | 1.0          | 8.9         | 2.0     |
| 3 | 8.05         | 600.0            | 7.0          | 6.3         | 9.0     |
| 4 | NaN          | 421.0            | 6.0          | 6.9         | 4.0     |

|   | bugs_reported | ai_usage_hours | cognitive_load | task_success |
|---|---------------|----------------|----------------|--------------|
| 0 | 1.0           | 0.71           | 5.4            | 1            |
| 1 | 3.0           | 1.75           | 4.7            | 1            |
| 2 | 0.0           | 2.27           | 2.2            | 1            |
| 3 | 5.0           | 1.40           | 5.9            | 0            |
| 4 | 0.0           | 1.26           | 6.3            | 1            |

The datasets were cleaned and readied before analysis in order to achieve accuracy and reliability. The identification and elimination of missing values were done to prevent the erroneous outcomes. Data quality was ensured by ensuring that duplicate records were eliminated.

The sentiment labels of the primary dataset including good and bad were then translated into standard categories of Positive and Negative in order to be consistent. There was also the cleaning of text data; the data was stripped off of unwarranted characters and symbols.

```

# data cleaning
df1['labels'] = df1['labels'].replace({
    'good' 'positive': 'Positive',
    'bad' 'negative': 'Negative'
})

df1.isnull().sum()

```

```

Unnamed: 0      1
tweets          17
labels          1
dtype: int64

df1 = df1.dropna()
df1.isnull().sum()

Unnamed: 0      0
tweets          0
labels          0
dtype: int64

```

In the secondary data, the variables like coding hours, the hours of AI usage, and other variables were numerically verified and structured accordingly. These preprocesses to the data were necessary to prepare the data to be clean, consistent, and useful to analyze it.

```

df2.isnull().sum()

hours_coding      2
coffee_intake_mg  5
distractions       1
sleep_hours       4
commits            2
bugs_reported      1
ai_usage_hours     2
cognitive_load     1
task_success       0
dtype: int64

df2 = df2.dropna()
df2.isnull().sum()

hours_coding      0
coffee_intake_mg  0
distractions       0
sleep_hours       0
commits            0
bugs_reported      0
ai_usage_hours     0
cognitive_load     0
task_success       0
dtype: int64

```

## Visual 1: Sentiment Distribution (Primary Dataset)

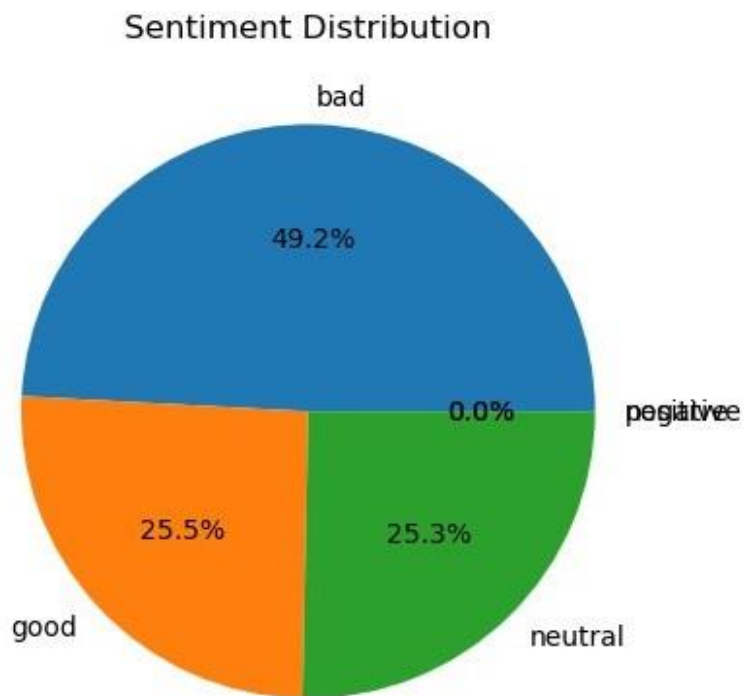
```

import matplotlib.pyplot as plt

df1['labels'].value_counts().plot(kind='pie', autopct='%1.1f%%')

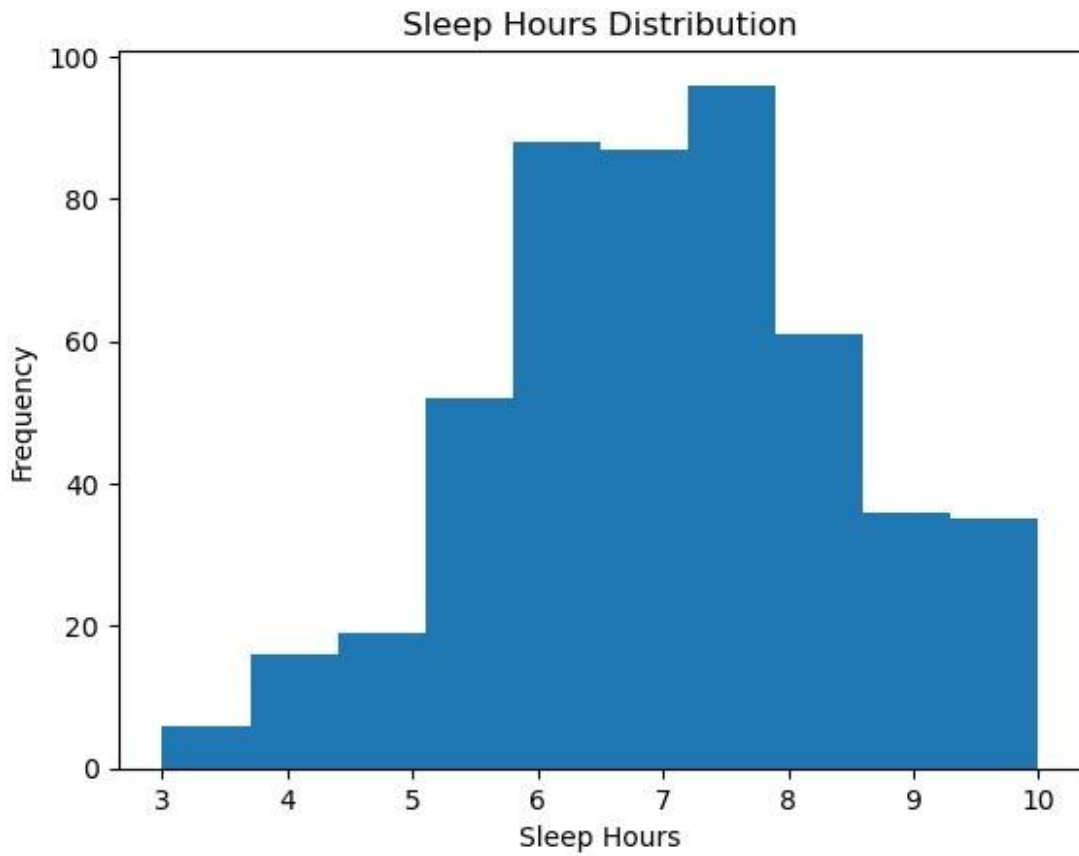
```

```
plt.title("Sentiment Distribution")
plt.ylabel("") # removes extra label
plt.show()
```



## Visual 2: Sleep Hours Distribution (Secondary Dataset)

```
df2['sleep_hours'].plot(kind='hist')
plt.title("Sleep Hours Distribution")
plt.xlabel("Sleep Hours")
plt.ylabel("Frequency")
plt.show()
```



## Visual 3

This graph demonstrates how the number of categories of sentiments (positive, negative, neutral) is distributed, as well as the mean level of productivity. The bars are used to display the number of tweets per sentiment and these aid us in knowing the number of users whose sentiment is negative, positive, or neutral towards ChatGPT. The horizontal line depicts the mean productivity level as obtained in the secondary data.

We can compare user opinions with total productivity by juxtaposing both in a single graph. This assists us in knowing whether positive sentiment is related to better performance. To illustrate, assuming that the majority of tweets are positive, and the average productivity is also high, then it would mean that those users possessing a good opinion about ChatGPT can get a better benefit of it. This chart gives a straightforward and easy means of relating the user perception and actual productivity results in the real world.

```
import pandas as pd

# Sentiment count
sentiment_counts = df1['labels'].value_counts().reset_index()
sentiment_counts.columns = ['Sentiment', 'Count']

# Add average productivity
avg_productivity = df2['task_success'].mean()
```

```
# Add same productivity value to all rows
sentiment_counts['Avg_Productivity'] = avg_productivity
```

```
print(sentiment_counts)
```

|   | Sentiment | Count  | Avg_Productivity |
|---|-----------|--------|------------------|
| 0 | bad       | 107794 | 0.606            |
| 1 | good      | 56009  | 0.606            |
| 2 | neutral   | 55487  | 0.606            |
| 3 | negative  | 2      | 0.606            |
| 4 | positive  | 2      | 0.606            |

```
import matplotlib.pyplot as plt
```

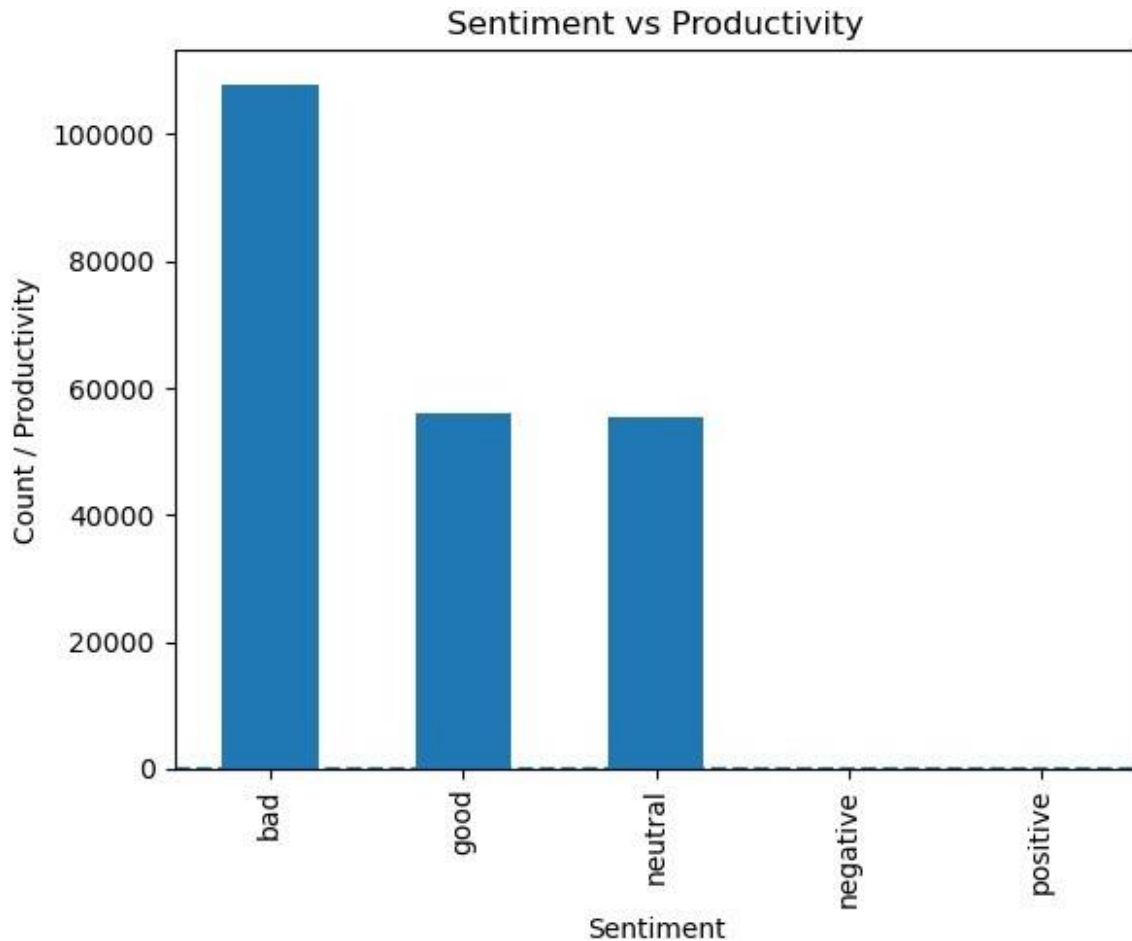
```
# Count sentiment
sentiment_counts = df1['labels'].value_counts()
```

```
# Average productivity
avg_productivity = df2['task_success'].mean()
```

```
# Bar chart (sentiment)
sentiment_counts.plot(kind='bar')
```

```
# Add line for productivity
plt.axhline(y=avg_productivity, linestyle='--')
```

```
plt.title("Sentiment vs Productivity")
plt.xlabel("Sentiment")
plt.ylabel("Count / Productivity")
plt.show()
```



## Findings

The evaluation reveals that the majority of users feel positive about ChatGPT, which means that the acceptance of AI tools is high. This implies that ChatGPT is helpful, useful, and easy to use by many users in their everyday activities. The presence of more positive feelings implies that individuals have faith in AI tools and are not afraid of using them in various aspects of life like education, writing code, and resolving problems.

Based on the secondary data, it was found that more frequent users of AI tools are more likely to be more active, including committing more and working more on productive tasks. This indicates that AI applications assist developers to work quickly, lessen efforts and enhance productivity. AI can assist developers to work faster and hence become more efficient.

Also, the trends in the duration of sleep and encoding behavior demonstrate that productivity is not determined by the use of AI only. To illustrate, good sleep and work hours are also good performance determinants. Nevertheless, the data shows clearly that AI utilization is relevant in enhancing productivity because the developers, who are more active users of AI tools, appear to be more productive than the less active ones.

## Patterns, Relationships, Trends, and Distributions

To understand the user behavior and productivity more in this analysis, we examined patterns, relationships, trends, and distributions in the two datasets. The sentiment data allowed us to see the tendencies in the opinion of the users, as we can see how people feel towards ChatGPT. The secondary data enabled us to see the trends in such variables as the usage of AI, the number of coding hours, or the number of commits. We also analyzed the associations among various variables, including the influence of the AI use on the activity of the developers. We also used distribution graphs to get an idea of the distribution of such values as the hours of sleep and productivity. On the whole, this discussion gave valuable information regarding the effect of AI tools on user behavior and performance.